```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from PIL import Image
import io

print("OpenCV version:", cv2.__version__)
print("Libraries imported successfully!")
```

```
OpenCV version: 5.0.0
Libraries imported successfully!
```

```
uploaded = files.upload()

filename = next(iter(uploaded))
image_bytes = np.frombuffer(uploaded[filename], np.uint8)

img = cv2.imdecode(image_bytes, cv2.IMREAD_COLOR)

if img is None:
    raise ValueError("Image could not be loaded. Please upload a valid image.")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully!")
print("Image dimensions:", gray.shape)
```

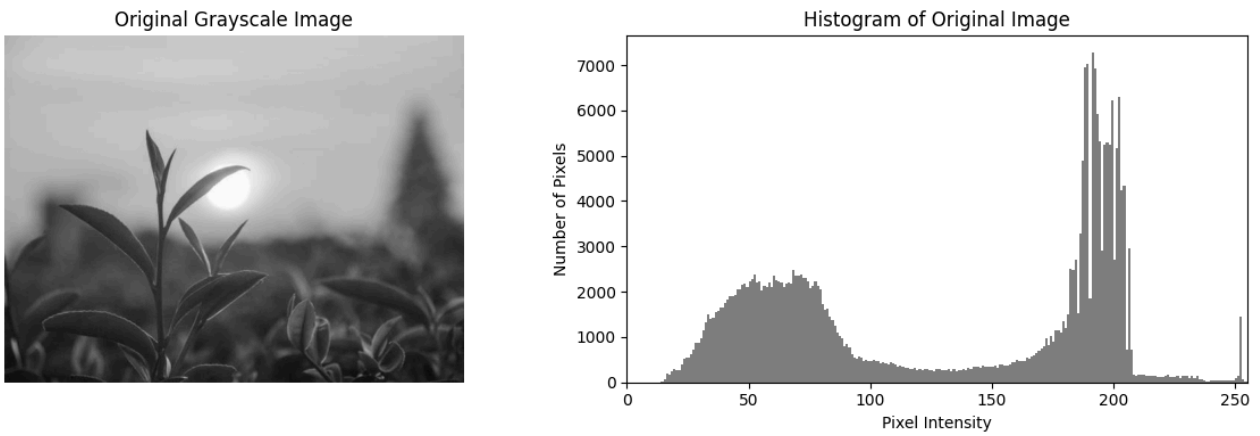
```
Choose files nature.jpg
nature.jpg(image/jpeg) - 27940 bytes, last modified: 13/10/2026 - 100% done
Saving nature.jpg to nature.jpg
Image loaded successfully!
Image dimensions: (460, 612)
```

```
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray", vmin=0, vmax=255)
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(gray.ravel(), bins=256, range=(0, 256), color="gray")
plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Number of Pixels")
plt.xlim(0, 255)

plt.tight_layout()
plt.show()
```



```
min_val = np.min(gray)
max_val = np.max(gray)

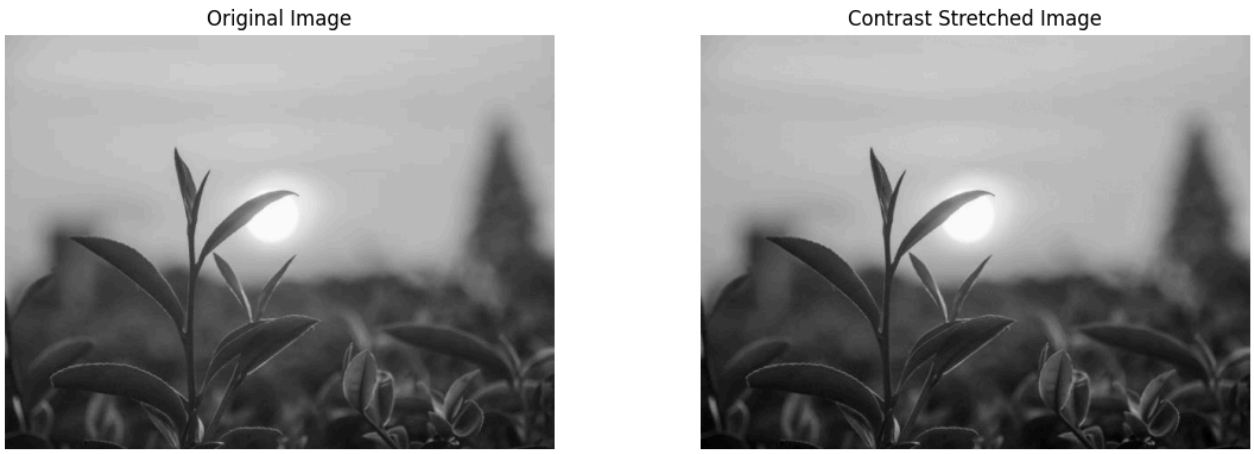
if max_val > min_val:
    stretched = cv2.normalize(
        gray, None, 0, 255, cv2.NORM_MINMAX
    )
else:
    stretched = gray.copy()

plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray", vmin=0, vmax=255)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(stretched, cmap="gray", vmin=0, vmax=255)
plt.title("Contrast Stretched Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```




```
equalized = cv2.equalizeHist(gray)

plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray", vmin=0, vmax=255)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(equalized, cmap="gray", vmin=0, vmax=255)
plt.title("Histogram Equalized Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```






```
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_img = clahe.apply(gray)

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray", vmin=0, vmax=255)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(equalized, cmap="gray", vmin=0, vmax=255)
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(clahe_img, cmap="gray", vmin=0, vmax=255)
plt.title("CLAHE Enhanced Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

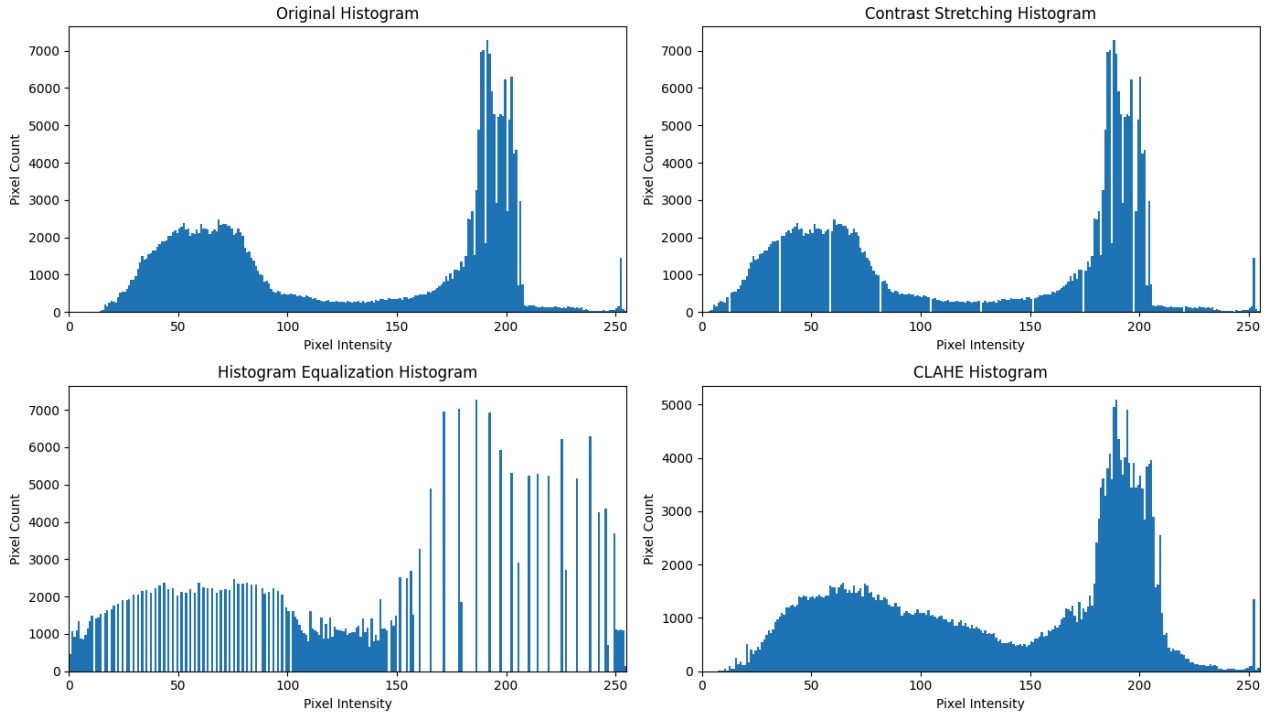



```
images = [
    ("Original", gray),
    ("Contrast Stretching", stretched),
    ("Histogram Equalization", equalized),
    ("CLAHE", clahe_img)
]

plt.figure(figsize=(14, 8))
```

```
for i, (name, image) in enumerate(images, 1):
    plt.subplot(2, 2, i)
    plt.hist(image.ravel(), bins=256, range=(0, 256))
    plt.title(name + " Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Pixel Count")
    plt.xlim(0, 255)

plt.tight_layout()
plt.show()
```




```
print(f"{'Method':<25} {'Mean':>10} {'Std Dev':>12} {'Min':>8} {'Max':>8}")
print("-" * 65)

for name, image in images:
    print(
        f"{name:<25} "
        f"{np.mean(image):>10.2f} "
        f"{np.std(image):>12.2f} "
        f"{np.min(image):>8} "
        f"{np.max(image):>8}"
    )
```

```
Method                          Mean      Std Dev      Min      Max
-----------------------------------------------------------------
Original                      128.53        66.39       11      255
Contrast Stretching           122.82        69.38        0      255
Histogram Equalization        128.83        74.16        0      255
CLAHE                         136.58        61.23        6      255
```

```
cv2.imwrite("contrast_stretched.png", stretched)
cv2.imwrite("histogram_equalized.png", equalized)
cv2.imwrite("clahe_enhanced.png", clahe_img)

print("Enhanced images saved successfully!")
```

```
files.download("contrast_stretched.png")
files.download("histogram_equalized.png")
files.download("clahe_enhanced.png")
```

```
Enhanced images saved successfully!
```

```
# Experiment No. 2
# Part 2: Questions and Answers

qa = [
("Q1. What is image contrast, and why is it important in image processing?",
"Image contrast is the difference between bright and dark regions of an image. It is important because it improves visil

("Q2. Explain the concept of an image histogram. What information does it provide?",
"An image histogram represents the distribution of pixel intensity values. For a grayscale image, values range from 0 (l

("Q3. Differentiate between Histogram Stretching and Histogram Equalization.",
"Histogram Stretching expands the existing intensity range, usually through a linear transformation. Histogram Equaliza

("Q4. What is CLAHE? How does it differ from standard Histogram Equalization?",
"CLAHE stands for Contrast Limited Adaptive Histogram Equalization. It enhances contrast in small image regions and lim

("Q5. Why is histogram equalization commonly applied before feature extraction and image segmentation?",
"It improves contrast and makes details, edges, and object boundaries more visible. This can help feature extraction an

("Q6. Mention three real-world applications where histogram equalization is widely used.",
"1. Medical imaging, such as X-rays and CT scans.\n2. Satellite image enhancement.\n3. Low-light photography."),

("Q7. What are the limitations of global histogram equalization?",
"It may amplify noise, over-enhance some regions, change the natural appearance of an image, and fail to improve detail

("Q8. How does contrast enhancement improve object detection and recognition systems?",
"It makes edges, textures, and object boundaries easier to distinguish. This can help computer vision systems extract fe

("Q9. Compare histogram-based enhancement techniques with brightness adjustment methods.",
"Brightness adjustment generally adds or subtracts an intensity value from pixels. Histogram-based methods redistribute

("Q10. Why is CLAHE preferred for medical imaging and low-light image enhancement?",
"CLAHE improves local contrast in areas with different illumination. Its contrast-limiting mechanism helps reduce exces
]

print("EXPERIMENT NO. 2")
print("CONTRAST ENHANCEMENT AND HISTOGRAM-BASED IMAGE PROCESSING")
print("=" * 65)

for question, answer in qa:
    print("\n" + question)
    print("-" * 65)
    print(answer)
    print()

print("RESULT: All 10 questions and answers displayed successfully.")
```

```
EXPERIMENT NO. 2
CONTRAST ENHANCEMENT AND HISTOGRAM-BASED IMAGE PROCESSING
=================================================================

Q1. What is image contrast, and why is it important in image processing?
-----------------------------------------------------------------
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-----------------------------------------------------------------
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```

```
Q6. Mention three real-world applications where histogram equalization is widely used.
--------------------------------------------------------------------
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2. Satellite image enhancement.
3. Low-light photography.


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--------------------------------------------------------------------
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RESULT: All 10 questions and answers displayed successfully.
```